

Summer 2025, 4100 Problem Set 4

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Use an orthogonal projection to find the point on the plane $S = \{\mathbf{x} \in \mathbb{R}^3 : 3x_1 - x_2 + 4x_3 = 0\}$ that is closest to the point $(-1, 4, 2)$.

2. Given vectors, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$, use Gram-Schmidt to find a corresponding orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ and show that $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{Span}\{\mathbf{q}_1, \mathbf{q}_2\}$.

3. Consider the Hilbert space $L^2(\mathbb{R}, w(x))$ where $w(x) = e^{-x^2}$, equipped with the inner product

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2}$$

and norm

$$\|f\| = \sqrt{\langle f, f \rangle} = \left(\int_{-\infty}^{\infty} |f(x)|^2 e^{-x^2} \right)^{1/2}.$$

Let $e_n = x^n$ denote the standard basis.

- a. Using the fact that $\int_{-\infty}^{\infty} e^{-x^2} = \sqrt{\pi}$, and using either integration by parts or parameter differentiation, show that $\langle e_i, e_j \rangle = \int_{-\infty}^{\infty} x^{i+j} e^{-x^2} = \begin{cases} \frac{(2n)! \sqrt{\pi}}{2^{2n} n!} & \text{if } i+j = 2n \\ 0 & \text{if } i+j = 2n+1 \end{cases}$

- b. Use the Gram-Schmidt and the result above to construct an orthonormal set of polynomials from the set $\{1, x, x^2\}$ and use it to find $g \in \mathbb{P}_2$ such that $f(1) = \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2}$ for all $f \in \mathbb{P}_2$.

4. Let $\{\mathbf{e}_1, \dots, \mathbf{e}_m\}$ be an orthonormal subset of $\mathbb{R}^n$. Prove that

$$\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{e}_1 \rangle^2 + \dots + \langle \mathbf{v}, \mathbf{e}_m \rangle^2$$

if and only if $\mathbf{v} \in \text{Span}(\mathbf{e}_1, \dots, \mathbf{e}_m)$

5. Let $Q \in \mathbb{R}^{n \times n}$ have orthonormal columns. Prove that,

- a. The rows of Q must be orthonormal as well

- b. $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^n$

6. Let $\mathbf{u} \in \mathbb{R}^n$ be such that $\|\mathbf{u}\| = 1$ and define the householder matrix, $H = I - 2\mathbf{u}\mathbf{u}^T$.

- a. Prove that H is symmetric and that $H = H^{-1}$.

- b. Prove that H acts as the identity on the hyperplane, $W = \{\mathbf{v} : \langle \mathbf{u}, \mathbf{v} \rangle = 0\}$

- c. Find a basis $\mathcal{B}$ for $\mathbb{R}^n$ such that $[H]_{\mathcal{B}}$ is diagonal.